TASK 09 – IMPLEMENT AND TEST THE CALCULATOR

1. Aim

To implement a complete milk payment calculator and test it using normal, boundary and invalid inputs. To verify the correctness of the calculator using test cases and assertions.

2. Introduction

The milk payment calculator calculates the payment for farmers based on the quantity of milk, fat percentage and SNF percentage.

The standard rate is ₹30.00 per litre at 3.5% fat and 8.5% SNF.

- Each 0.1% increase or decrease in fat changes the rate by ₹0.40.
- Each 0.1% increase or decrease in SNF changes the rate by ₹0.30.
- The final payment is rounded to two decimal places.

3. Testing Conditions

The calculator is tested using the following conditions:

A. Normal entries

- Fat and SNF values above the standard.
- Fat and SNF values below the standard.

B. Boundary entries

- Standard values: 3.5% fat and 8.5% SNF.
- Lowest allowed fat: 2.0%.
- Highest allowed fat: 7.0%.

C. Invalid entries

- Text instead of a number.
- Negative milk quantity.
- Fat below the allowed range.
- Fat above the allowed range.

4. Test Table

Test case	Input (Litres, Fat, SNF)	Expected result
Normal – above standard	8.5, 4.2, 8.7	₹283.90
Normal – below standard	12, 3.1, 8.3	₹333.60
Standard boundary	10, 3.5, 8.5	₹300.00

Lowest fat boundary	10, 2.0, 8.5	₹240.00
Highest fat boundary	10, 7.0, 8.5	₹440.00
Text input	abc, 3.5, 8.5	Invalid input
Negative quantity	-5, 3.5, 8.5	Invalid input
Fat below range	10, 1.5, 8.5	Invalid input
Fat above range	10, 7.5, 8.5	Invalid input

The actual results are checked by executing the Python program.

5. Five-Entry Manual Calculation

The following five entries are used to verify the ten-day total.

Entry	Litres	Fat	SNF	Rate/Litre	Payment
1	10	3.5	8.5	₹30.00	₹300.00
2	8.5	4.2	8.7	₹33.40	₹283.90
3	12	3.1	8.3	₹27.80	₹333.60
4	5	3.0	8.0	₹26.50	₹132.50
5	7.5	4.0	8.5	₹32.00	₹240.00

Manual total:

₹300.00 + ₹283.90 + ₹333.60 + ₹132.50 + ₹240.00 = **₹1,290.00**

The manually calculated total is compared with the total produced by the Python program.

6. Alternative Concept – Manual Testing vs Automated Testing

Manual testing

In manual testing, the user enters each test input separately and checks the output. Whenever the program changes, all the test cases must be entered and checked again.

Automated testing

In automated testing, test cases are stored in the program and checked using assert statements. The tests run automatically whenever the program is executed.

Comparison

Manual testing	Automated testing
Inputs are entered one by one.	Inputs are stored in test cases.
Results are checked manually.	Assertions check results automatically.

Repeating tests takes more time.	All tests can be rerun quickly.
Human checking may cause mistakes.	Gives consistent checks for expected results.

Automated testing is better when the calculator is modified frequently because all test cases can be executed again without manually entering each value.

7. Complexity Analysis

Let n be the number of test cases.

- Time complexity: O(n), because each test case is checked once.
- Extra space complexity: O(n), because the test cases are stored in a list.

8. Result

The milk payment calculator was implemented and tested using normal, boundary and invalid inputs. The five-entry total was verified manually as ₹1,290.00 and compared with the program resul